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The role of radiotherapy and chemotherapy in extraskeletal myxoid chondrosarcoma

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Abstract

Background Extraskeletal myxoid chondrosarcoma (EMC) is an extremely rare subtype of sarcoma characterized by NR4A3 gene rearrangement. Despite being considered slowly progressive sarcomas, EMCs tend to have local recurrences and distant metastases in the late stages. This study aimed to investigate the prognostic factors of EMCs, especially the effect of (neo)adjuvant radiotherapy or chemotherapy on localized EMCs and chemotherapy on advanced-stage EMCs.

Methods We retrospectively analyzed 171 patients pathologically diagnosed with EMCs between 2002 and 2022 using the Japanese National Bone and Soft Tissue Tumor Registry Database.

Results Disease-specific survival was significantly shorter in the group with distant metastasis at presentation ($n = 29$) than in the group without ($n = 142$) (5-year disease-specific survival, 75.8% [95% CI: 54.7–89.1] vs. 91.3% [95% CI: 83.0–95.8]; $p = 0.012$). Multivariate analysis showed that an R1 or R2 surgical margin was a risk factor for unfavorable local recurrence (hazard ratio [HR] 4.76 [95% CI: 1.72–13.15]; $p = 0.003$). No association was found between (neo)adjuvant radiotherapy and local recurrence rates. With respect to disease-specific survival, the tumor site of the trunk (HR 6.28 [95% CI: 1.30–30.49]; $p = 0.023$) and larger size (HR 1.18 [95% CI: 1.06–1.33]; $p = 0.004$) were risk factors for unfavorable disease-specific survival. No association was found between (neo)adjuvant, or advanced-stage chemotherapy and disease-specific survival rates.

Conclusions Patients with distant metastases at presentation had significantly shorter survival rates than those without. Wide resection is mandatory to reduce the risk of local recurrence of localized EMCs, and the local control effect of (neo)adjuvant radiotherapy is limited. Chemotherapy has a limited effect on improving survival.

Keywords Extraskeletal, Myxoid, Chondrosarcoma, Surgery, Adjuvant, Radiotherapy, Chemotherapy, Prognosis

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Background

Extraskelletal myxoid chondrosarcoma (EMC) is an ultra-rare sarcoma subtype that accounts for approximately 1% of all soft-tissue sarcomas in adults [1, 2]. The median age of onset is approximately 60 years, and the male-to-female ratio is 2:1 [1, 2]. Most EMCs occur in the proximal extremities, with the thigh being the most common site [1, 2]. They can also occur in the trunk, head and neck, paraspinal muscles, abdomen, retroperitoneal cavity [3], and bone [4–6]. Originally thought to be cartilaginous tumors, EMCs are now classified as mesenchymal tumors with uncertain differentiation [7]. Indeed, these tumors are characterized by a rearrangement of the NR4A3 gene [7] and do not undergo chondrogenic differentiation, despite their name, which has been retained only for historical reasons [7, 8].

Similar to other soft-tissue sarcomas, EMCs usually metastasize to the lungs, although extrapulmonary metastasis can also occur [9, 10]. Despite being considered slowly progressive sarcomas, EMCs tend to recur locally and metastasize in the late stages [11]. The long-term risk of metastasis is high [11]. Nevertheless, the progression of distant metastasis is gradual, allowing patients to survive for several years without intervention. The 5- and 10-year local recurrence-free survival rates have been reported to be 41–94% [12–14] and 31–90% [13, 14], respectively, and the 5- and 10-year metastasis-free survival rates are 64–73% [9, 14] and 35–69% [9, 14], respectively. Moreover, the 5- and 10-year overall survival rates are 71–100% [2, 3, 9, 12–15] and 60–88% [2, 3, 9, 13–15], respectively. The frequency of distant metastases at the initial diagnosis is 15–23% [2, 16].

The treatment of localized EMCs is based on wide resection; however, the efficacy of (neo)adjuvant radiotherapy is controversial because, like conventional chondrosarcomas, EMCs have long been considered refractory to radiotherapy. Studies on (neo)adjuvant radiotherapy are scarce, and their efficacy remains controversial [1, 2, 9, 12–14, 17]. Furthermore, no study has investigated the efficacy of (neo)adjuvant chemotherapy in these patients [18].

Similar to conventional chondrosarcomas, advanced-stage EMCs are considered resistant to the cytotoxic agents commonly used in the treatment of sarcomas. In older retrospective studies, there are few reports on responses to regimens containing anthracycline, dacarbazine, or ifosfamide [9, 13, 19–21]. However, in a recent retrospective study in which EMCs were molecularly confirmed, a 40% response rate and a median 8-month progression-free survival period were reported with an anthracycline-based regimen [22]. In a prospective study, a 19-month progression-free survival period was reported with pazopanib [23], and a 12-month

progression-free survival period was reported with trabectedin [24].

This study aimed to investigate the prognostic factors for EMCs, particularly the effects of (neo)adjuvant radiotherapy and chemotherapy on localized EMCs and the effects of chemotherapy on advanced-stage EMCs.

Methods

Initially, 223 patients pathologically diagnosed with EMC between January 2002 and December 2022 were selected from the Japanese National Bone and Soft Tissue Tumor Registry Database [25]. After excluding 18 patients with unknown tumor size, 12 with unknown histological grade, and 22 with a follow-up period of less than 6 months after diagnosis, the remaining 171 patients were retrospectively analyzed. The registry was approved by the Institutional Review Board (IRB) of the National Cancer Center, and the study using the registry was approved by the Musculoskeletal Tumor Committee of the Japanese Orthopaedic Association. The IRB waived the requirement for informed consent because the data extracted from the database were anonymized.

The following information was obtained from the database: age, sex, previous surgery, site of origin, depth, tumor size, presence of regional lymph node metastasis at presentation, presence of distant metastasis at presentation, histological grade [26], biopsy method, and surgical margins (microscopically negative margins were defined as [R0], macroscopically negative but microscopically positive margins as [R1], and macroscopically positive margins as [R2]) [27]. In addition, (neo)adjuvant radiotherapy, (neo)adjuvant chemotherapy, local recurrence, distant metastasis, disease-specific survival, and follow-up period were extracted. Tumor locations were grouped into upper extremities, lower extremities, and trunk.

Regarding perioperative radiotherapy for localized EMCs, neoadjuvant radiotherapy was administered at a total dose of 40–50 Gy in 2 Gy fractions in four patients, and adjuvant radiotherapy at a total dose of 50–66 Gy in 2 Gy fractions in 20 patients. Neoadjuvant or adjuvant radiotherapy is administered to prevent local recurrence in patients with close surgical margins. Regarding perioperative chemotherapy for localized EMCs, five patients received 3–6 cycles of adriamycin 60 mg/m², plus ifosfamide 10 g/m² [28], and one patient received 3–6 cycles of adriamycin monotherapy 75 mg/m². Neoadjuvant chemotherapy was administered to four patients, adjuvant chemotherapy to one patient, and both neoadjuvant and adjuvant chemotherapy to one patient.

To evaluate any association between two variables, we used either the chi-square test or Fisher's exact test, while the Mann–Whitney U test was used for nonparametric statistical analysis of the differences between two

independent samples. Local recurrence-free survival was defined as the interval between surgery and local recurrence or the last follow-up. Distant metastasis-free survival was defined as the interval between diagnosis and distant metastasis or the last follow-up. Disease-specific survival was defined as the interval between diagnosis and death from the disease or the last follow-up. Univariate analysis was performed using Cox proportional hazards regression and Kaplan–Meier survival analyses (log-rank test) to examine the association between each variable and local recurrence-free survival, distant metastasis-free survival, and disease-specific survival. Multivariate analyses included stepwise Cox proportional hazards regression analysis and multivariate logistic regression analysis of all variables. Statistical significance was set at $p < 0.05$. Analyses were performed using JMP 18 software (SAS Institute Inc., Cary, NC, USA) and IBM SPSS (version 28.0; IBM Co., Armonk, NY, USA).

Results

A total of 102 (59.6%) males and 69 (40.4%) females with a median age of 62 years (interquartile range [IQR], 48–72 years) were included in the study. The tumor location was the lower extremity in 100 patients, the upper extremity in 16, and the trunk in 55 patients. The median tumor size was 8 cm (IQR, 5–11.5 cm). Of the 171 patients, 142 (83.0%) had no distant metastases at diagnosis, and the remaining 29 (17.0%) had distant metastases at diagnosis. Nineteen patients (11.1%) died from the tumors, and the median time from the date of diagnosis to death due to the tumor was 33 months (IQR, 10–82 months). Disease-specific survival rates at 1, 2, and 5 years were 96.4%, 95.0%, and 88.6%, respectively. The median disease-specific survival and follow-up periods were 38 months (IQR, 23–71). Univariate analysis showed significantly shorter survival in the group with distant metastases at initial presentation than in the group without (5-year disease-specific survival, 75.8% [95% CI: 54.7–89.1] vs. 91.3% [95% CI: 83.0–95.8]; $p = 0.012$; Table 1; Fig. 1).

Patients without metastases at diagnosis

A total of 142 patients had no distant metastases at the time of initial diagnosis. Regional lymph node metastasis was present in three patients (2.1%) at the time of diagnosis. Regarding surgery, 104 (73.2%) patients underwent R0 resection, 22 (15.5%) underwent R1 resection, eight (5.6%) underwent R2 resection, and eight (5.6%) did not undergo surgery. Of the eight patients who did not undergo surgery, two received carbon ion therapy, one received proton beam therapy, and one received conventional radiotherapy. Of the eight patients who did not undergo surgery, three received chemotherapy for advanced stages.

For the prognostic analysis, eight patients who did not undergo surgery were excluded, and the remaining 134 patients were included. Local recurrence occurred in 16 patients (11.9%), and the median time from surgery to local recurrence was 15 months (IQR, 4.5–63.5). Distant metastases occurred in 39 patients (29.1%), and the median time from diagnosis to distant metastasis was 16 months (IQR, 10–31 months). Nine patients (6.7%) died due to the tumor, and the median time from diagnosis to death due to the tumor was 36 months (IQR, 25–69 months).

For local recurrence, univariate analysis showed that an R1 or R2 surgical margin was a risk factor for unfavorable local recurrence (HR 4.76 [95% CI: 1.72–13.15]; $p = 0.003$; Table 2). There was no association between (neo)adjuvant radiotherapy and local recurrence rate (HR 0.50 [95% CI: 0.11–2.25]; $p = 0.365$; Table 2; Fig. 2). Multivariate analysis showed that an R1 or R2 surgical margin was an independent risk factor for unfavorable local recurrence (HR 4.76 [95% CI: 1.72–13.15]; $p = 0.003$; Table 2). Of the 24 patients who received (neo)adjuvant radiotherapy, 10 (41.7%) had R1 or R2 surgical margins, whereas only 20 (18.2%) of the 110 patients who did not receive (neo)adjuvant radiotherapy had R1 or R2 surgical margins. Of the 24 patients who received (neo)adjuvant radiotherapy, two (8.3%) experienced local recurrence, and of the 110 patients who did not receive (neo)adjuvant radiotherapy, 14 (12.7%) experienced local recurrence.

For distant metastases, univariate analysis showed that larger size [HR 1.12 (95% CI: 1.05–1.18); $p < 0.001$] and R1 or R2 surgical margins [HR 2.37 (95% CI: 1.21–4.64); $p = 0.012$] were risk factors for unfavorable distant metastasis (Table 3). There was no association between (neo)adjuvant chemotherapy and the distant metastasis rate (HR 0.57 [95% CI: 0.08–4.19]; $p = 0.584$; Table 3). Multivariate analysis showed that larger tumor size (hazard ratio (HR) 1.10 [95% confidence interval (CI): 1.04–1.17]; $p = 0.002$), high histological grade (HR 2.55 [95% CI: 1.18–5.52]; $p = 0.018$), and R1 or R2 surgical margins (HR 3.40 [95% CI: 1.57–7.38]; $p = 0.002$) were independent risk factors for distant metastasis (Table 3). Among the six patients who received (neo)adjuvant chemotherapy, one (16.7%) had distant metastasis, and among the 128 patients who did not receive (neo)adjuvant chemotherapy, 38 (29.7%) had distant metastasis.

Regarding disease-specific survival, univariate analysis revealed that tumor location on the trunk (HR 6.17 [95% CI: 1.28–29.75]; $p = 0.023$), larger tumor size (HR 1.20 [95% CI: 1.06–1.36]; $p = 0.003$), and (neo)adjuvant radiotherapy (HR 5.05 [95% CI: 1.34–19.04]; $p = 0.017$) were identified as risk factors for unfavorable disease-specific survival (Table 4). There was no association between (neo)adjuvant chemotherapy and disease-specific survival (HR 3.71 [95% CI: 0.44–31.02]; $p = 0.226$; Table 4).

Table 1 Patient characteristics and clinical outcomes between groups with and without distant metastasis at diagnosis

Variable	Total patients (N = 171)	Metastases at diagnosis		p-Value
		No (N = 142)	Yes (N = 29)	
Age (years)				
Median (IQR)	62 (48–72)	62.5 (47.8–72)	59 (48–72.5)	0.829
Sex, n (%)				0.124
Male	102 (59.7)	81 (57.0)	21 (72.4)	
Female	69 (40.3)	61 (43.0)	8 (27.6)	
Previous surgery, n (%)				0.363 [#]
No	153 (89.5)	128 (90.1)	25 (86.2)	
Yes	18 (10.5)	14 (9.9)	4 (13.8)	
Tumor site, n (%)				0.446 ^a
Lower limb	100 (58.5)	81 (57.0)	19 (65.5)	
thigh	61 (35.7)	49 (34.5)	12 (41.4)	
knee	17 (9.9)	14 (9.9)	3 (10.3)	
lower leg	8 (4.7)	7 (4.9)	1 (3.5)	
ankle	4 (2.3)	4 (2.8)	0	
foot	10 (5.9)	7 (4.9)	3 (10.3)	
Upper limb	16 (9.4)	15 (10.6)	1 (3.5)	
upper arm	6 (3.5)	5 (3.5)	1 (3.5)	
elbow	3 (1.8)	3 (2.1)	0	
forearm	2 (1.2)	2 (1.4)	0	
hand	5 (2.9)	5 (3.5)	0	
Trunk	55 (32.2)	46 (32.4)	9 (31.0)	
perineum	1 (0.6)	1 (0.7)	0	
intrathoracic	1 (0.6)	1 (0.7)	0	
chest wall	5 (2.9)	4 (2.8)	1 (3.5)	
neck	1 (0.6)	1 (0.7)	0	
shoulder	9 (5.3)	8 (5.6)	1 (3.5)	
retroperitoneum	3 (1.8)	3 (2.1)	0	
lumbar	1 (0.6)	1 (0.7)	0	
groin	8 (4.7)	8 (5.6)	0	
back	4 (2.3)	3 (2.1)	1 (3.5)	
abdominal wall	4 (2.3)	3 (2.1)	1 (3.5)	
axilla	3 (1.8)	3 (2.1)	0	
buttock	15 (8.8)	10 (7.0)	5 (17.2)	
Tumor depth, n (%)				0.945 [#]
Superficial	24 (14.0)	22 (15.5)	2 (6.9)	
Deep	147 (86.0)	120 (84.5)	27 (93.1)	
Tumor size (cm)				
Median (IQR)	8 (5–11.5)	7.9 (4.9–11)	8.6 (5.4–13.9)	0.135
Histological grade, n (%)				0.051
Low	69 (40.4)	62 (43.7)	7 (24.1)	
High	102 (59.7)	80 (56.3)	22 (75.9)	
Biopsy, n (%)				
Needle	17 (9.9)	15 (10.6)	2 (6.9)	
Open	125 (73.1)	99 (69.7)	26 (89.7)	
Excisional	10 (5.8)	10 (7.0)	0	
None	19 (11.1)	18 (12.7)	1 (3.4)	
Surgical margin, n (%)				
R0	117 (68.4)	104 (73.2)	13 (44.8)	
R1	30 (17.5)	22 (15.5)	8 (27.6)	
R2	9 (5.3)	8 (5.6)	1 (3.4)	
No surgery	15 (8.8)	8 (5.6)	7 (24.1)	
Radiotherapy, n (%)				

Table 1 (continued)

Variable	Total patients (N= 171)	Metastases at diagnosis		p-Value
		No (N= 142)	Yes (N= 29)	
Neoadjuvant	5 (2.9)	4 (2.8)	1 (3.4)	
Adjuvant	23 (13.5)	20 (14.1)	3 (10.3)	
Palliative	8 (4.7)	5 (3.5)	3 (10.3)	
None	129 (75.4)	107 (75.4)	22 (75.9)	
Chemotherapy, n (%)				
Neoadjuvant	7 (4.1)	5 (3.5)	0	
Adjuvant	4 (2.3)	4 (2.8)	0	
Advanced disease	30 (17.5)	15 (10.6)	16 (55.2)	
None	133 (77.8)	120 (84.5)	13 (44.8)	
Died of disease, n (%)				< 0.001*
No	152 (88.9)	132 (93.0)	20 (69.0)	
Yes	19 (11.1)	10 (7.0)	9 (31.0)	
Five-year disease-specific survival, % (95% CI)	88.6 (81.1–93.3)	91.3 (83.0–95.8)	75.8 (54.7–89.1)	0.012*
Follow-up period after diagnosis (months), median (IQR)	38 (23–71)	38 (25.8–71)	41 (14.5–79)	0.987

IQR, interquartile range; CI, confidence interval; * Statistically significant; # Fisher exact test was used; (a) comparison in major classification (lower limb, upper limb, and trunk)

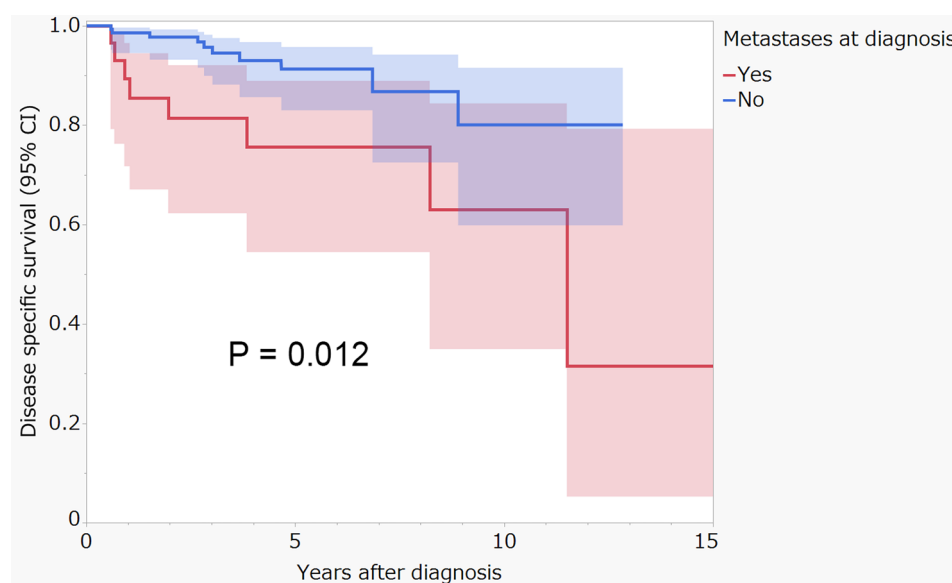


Fig. 1 Disease-specific survival rates of patients with and without distant metastases at diagnosis. The shading around the curves represents the 95% confidence interval

Multivariate analysis showed that tumor location on the trunk (HR 6.28; 95% CI: 1.30–30.49; $p=0.023$) and larger tumor size (HR 1.18; 95% CI: 1.06–1.33; $p=0.004$) were identified as independent risk factors for unfavorable disease-specific survival (Table 4). Of the six patients who received (neo)adjuvant chemotherapy, four (66.7%) were tumor-free survivors, one (16.7%) survived with the tumor, and one (16.7%) died from the tumor. Of the 128 patients who did not receive (neo)adjuvant chemotherapy, 88 (68.8%) were tumor-free survivors, 28 (21.9%) survived with tumors, eight (6.3%) died from tumors, and four (3.1%) died from other causes.

Next, we investigated the risk factors for death due to tumors using multivariate logistic regression analysis (stepwise method). We did this by adding local recurrence and distant metastasis as variables to all the variables in Table 4. The results showed that the risk factors for death due to tumors were tumor location on the trunk (odds ratio (OR) 8.52 [95% CI: 1.50–48.36]; $p=0.016$) and the occurrence of distant metastasis (OR 26.26 [95% CI: 2.99–231.04]; $p=0.003$; Table S1). No association was found between local recurrence and tumor mortality.

Table 2 Univariate Cox regression analysis of local recurrence-free survival in patients who had localized extraskeletal myxoid chondrosarcoma and underwent surgery

Univariate analysis			
Variable (n = 134)	No. of patients	Hazard ratio (95% CI)	p-value
Age (years)	134	1.02 (0.99–1.06)	0.151
Sex			
Male	76	1	
Female	58	1.34 (0.48–3.73)	0.575
Previous surgery			
No	121	1	
Yes	13	1.41 (0.32–6.27)	0.649
Site			
Lower limb	79	1	
Upper limb	14	1.84 (0.37–9.15)	0.454
Trunk	41	2.12 (0.71–6.31)	0.177
Tumor depth			
Superficial	22	1	
Deep	112	1.87 (0.41–8.48)	0.419
Tumor size (cm)	134	1.05 (0.94–1.15)	0.334
Histological grade			
Low	57	1	
High	77	1.13 (0.40–3.17)	0.820
Surgical margin			
R0	104	1	
R1 or R2	30	4.76 (1.72–13.15)	0.003*
(Neo)adjuvant radiotherapy			
No	110	1	
Yes	24	0.50 (0.11–2.25)	0.365
(Neo)adjuvant chemotherapy			
No	128	1	
Yes	6	1.90 (0.25–14.64)	0.538
Multivariate analysis			
Variable (n = 134)	No. of patients	Hazard ratio (95% CI)	p-value
Surgical margin			
R0	104	1	
R1 or R2	30	4.76 (1.72–13.15)	0.003*

*Statistically significant. CI, confidence interval

Patients with metastases at diagnosis

A total of 29 patients had distant metastases at the time of diagnosis. Of these, 27 patients had lung metastases, and two had peritoneal dissemination. Regional lymph node metastasis was present in six patients (20.7%) at the time of initial diagnosis. Regarding surgery for the primary tumor, 13 patients (44.8%) underwent R0 resection, eight (27.6%) underwent R1 resection, one (3.4%) underwent R2 resection, and seven (24.1%) did not undergo surgery. One patient received neoadjuvant radiotherapy, three received adjuvant radiotherapy, one received palliative radiotherapy for the primary tumor, and two received palliative radiotherapy for metastatic lesions. A total of 16 patients (55.2%) received chemotherapy for advanced disease: seven received adriamycin-ifosfamide, two received adriamycin monotherapy, two received adriamycin-cisplatin, one received ifosfamide-etoposide,

one received gemcitabine-docetaxel, one received eribulin, four received pazopanib, and two received trabectedin. Eight patients (27.6%) underwent metastasectomy, including six, one, and one who underwent lung, bone, and lymph node resections, respectively.

Among the 16 patients who received advanced-stage chemotherapy, one (6.3%) survived tumor-free, nine (56.3%) survived with tumors, five (31.3%) died of the tumors, and one (6.3%) died of other causes. Among the 13 patients who did not receive advanced-stage chemotherapy, nine (69.2%) survived with tumors, and four (30.8%) died of tumors. Univariate analysis showed no significant difference in disease-specific survival between the group that received advanced-stage chemotherapy and the group that did not (5-year disease-specific survival: 72.7% [95% CI: 45.3–89.5] vs. 78.6% [95% CI: 42.0–94.9]; $p = 0.276$; Fig. 3). The median disease specific

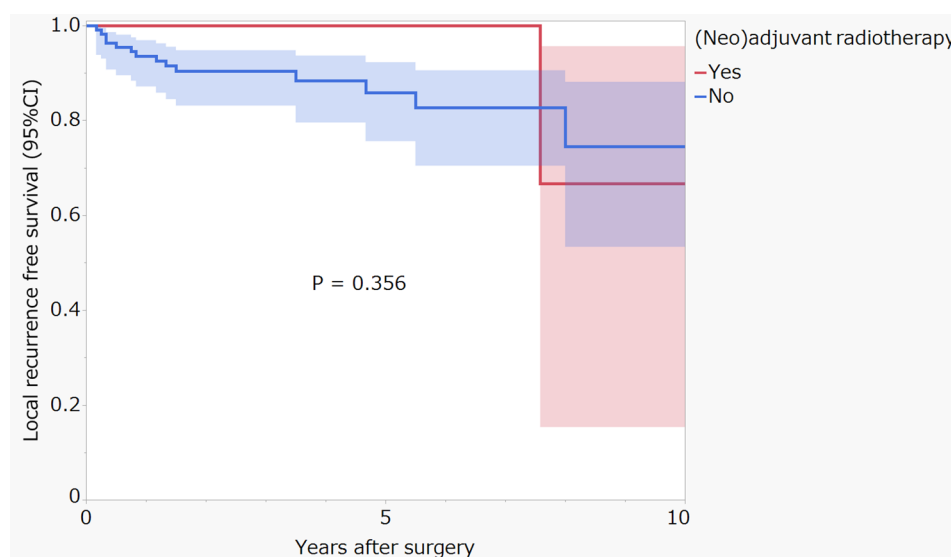


Fig. 2 Local recurrence-free survival rates of patients with and without (neo)adjuvant radiotherapy. The shading around the curves represents the 95% confidence interval

survival and follow-up periods were both 41 months (IQR, 14.5–79).

Discussion

In this study, survival was significantly shorter in the group with distant metastases at initial presentation, which is consistent with previous reports [1, 2, 13, 16, 21]. In localized EMCs, the risk factors for local recurrence were inadequate surgical margins, and the local control effect of (neo)adjuvant radiotherapy was limited. The risk factors for distant metastasis were large tumor size, high histological grade, and inadequate surgical margins, whereas the poor prognostic factors for survival were trunk origin and large tumor size. The occurrence of distant metastases, but not local recurrence, is associated with an increased risk of tumor-related death. The efficacy of (neo)adjuvant chemotherapy in suppressing distant metastases and improving survival is limited. Furthermore, the survival-prolonging effects of chemotherapy in advanced-stage EMC are limited. The reported poor survival prognostic factors were older age [1–3], larger tumor size [1–3, 21, 29], trunk location [3], high histological grade [1, 2], unresectable primary tumor [2], and presence of distant metastases at initial presentation [13, 16].

Localized extraskeletal myxoid chondrosarcoma

In the present study, patients who underwent surgery with R1 or R2 surgical margins had an increased risk of local recurrence. Similar to localized soft-tissue sarcomas, wide resection with negative microscopic margins is the standard of care for localized EMCs [30]. However, the local recurrence risk after wide resection is higher

than that after wide resection for other soft tissue sarcomas, ranging from 6 to 59% at 5 years [12–14]. Risk factors for local recurrence have been reported to be positive surgical margins [13, 15, 31, 32] and a truncal location [13, 21, 29].

In this study, no association was found between (neo)adjuvant radiotherapy and local recurrence rate. Drilon et al. [13] reported no difference in local recurrence risk between groups that received surgery and (neo)adjuvant radiotherapy and those that received surgery alone (41 vs. 35%, $p=0.79$). Other studies have reported similar results [1, 2, 16]. However, a retrospective study of 41 patients with EMC at a large tertiary center showed better local control with surgery and radiotherapy than with surgery alone (10-year local control rate, 100% vs. 63%; $p=0.004$) [14]. Another multicenter retrospective study involving only localized and molecularly confirmed cases showed a trend toward better local control in the group that underwent surgery and (neo)adjuvant radiotherapy than in the surgery-alone group (local recurrence rate: 1/10 (10%) vs. 7/17 (41%), $p=0.08$) [9]. This suggests that (neo)adjuvant radiotherapy and surgical resection may be considered for localized disease, especially when tumor shrinkage is required for the surgical downstaging of larger lesions [30].

In our study, patients with larger tumors, higher histological grades, and R1 or R2 surgical margins had an increased risk of distant metastasis. Previous studies have similarly found an increased rate of distant metastasis in patients with larger tumors [9, 32]. In our study, no association was observed between (neo)adjuvant chemotherapy and distant metastasis rates. Despite the high risk of distant metastasis after radical surgery, no studies

Table 3 Univariate Cox regression analysis of distant metastasis-free survival in patients who had localized extraskeletal myxoid chondrosarcoma and underwent surgery

Univariate analysis			
Variable (n = 134)	No. of patients	Hazard ratio (95% CI)	p-value
Age (years)	134	1.01 (0.99–1.04)	0.188
Sex			
Male	76	1	
Female	58	0.81 (0.43–1.55)	0.525
Previous surgery			
No	121	1	
Yes	13	0.21 (0.03–1.50)	0.119
Site			
Lower limb	79	1	
Upper limb	14	1.01 (0.35–2.95)	0.986
Trunk	41	1.29 (0.66–2.55)	0.455
Tumor depth			
Superficial	22	1	
Deep	112	1.33 (0.52–3.40)	0.555
Tumor size (cm)	134	1.12 (1.05–1.18)	<0.001*
Histological grade			
Low	57	1	
High	77	1.81 (0.92–3.58)	0.087
Surgical margin			
R0	104	1	
R1 or R2	30	2.37 (1.21–4.64)	0.012*
(Neo)adjuvant radiotherapy			
No	110	1	
Yes	24	1.76 (0.86–3.61)	0.125
(Neo)adjuvant chemotherapy			
No	128	1	
Yes	6	0.57 (0.08–4.19)	0.584
Multivariate analysis			
Variable (n = 134)	No. of patients	Hazard ratio (95% CI)	p-value
Tumor size (cm)	134	1.10 (1.04–1.17)	0.002*
Histological grade			
Low	57	1	
High	77	2.55 (1.18–5.52)	0.018*
Surgical margin			
R0	104	1	
R1 or R2	30	3.40 (1.57–7.38)	0.002*

*Statistically significant. CI, confidence interval

have supported the use of (neo)adjuvant chemotherapy in EMC [1, 2, 12–15]. The risk of distant metastasis is significantly higher in EMC than in skeletal ones [4]. The percentages of translocation types were 80% for NR4A3-EWS, 16% for NR4A3-TAF15, 2% for NR4A3-TCF12, and 2% for NR4A3-TFG [9]. The risk of distant metastasis was lower in patients with NR4A3-EWS translocations than in those with NR4A3-TAF15 translocations [9].

In this study, mortality increased in patients whose tumors were located on the trunk and were larger. Brown et al. [2] found that tumors at axial locations had the worst OS, those on the pelvis had a 5-year survival rate of 67.2%, and those at truncal locations had a

5-year survival rate of 66.0%. In 2016, an analysis of the Surveillance, Epidemiology, and End Results database examined the outcomes of 156 patients diagnosed with localized EMC between 2004 and 2012 [17]. Of these patients, 94% underwent surgery and 32% received radiotherapy (29% underwent combined surgical resection and radiotherapy). At a median follow-up of 33 months, the cancer-specific survival rates at 3 and 5 years favored radiotherapy (97% vs. 85% and 94% vs. 85%, respectively; $p=0.01$) [17]. In our study, no association was observed between (neo)adjuvant chemotherapy and disease-specific survival.

Table 4 Univariate Cox regression analysis of disease-specific survival in patients who had localized extraskeletal myxoid chondrosarcoma and underwent surgery

Univariate analysis			
Variable (n = 134)	No. of patients	Hazard ratio (95% CI)	p-value
Age (years)	134	1.05 (0.99–1.11)	0.111
Sex			
Male	76	1	
Female	58	0.80 (0.20–3.28)	0.756
Previous surgery			
No	121	1	
Yes	13	0	0.999
Site			
Lower limb	79	1	
Upper limb	14	0	0.999
Trunk	41	6.17 (1.28–29.75)	0.023*
Tumor depth			
Superficial	22	1	
Deep	112	0.67 (0.14–3.30)	0.627
Tumor size (cm)	134	1.20 (1.06–1.36)	0.003*
Histological grade			
Low	57	1	
High	77	2.60 (0.54–12.63)	0.236
Surgical margin			
R0	104	1	
R1 or R2	30	3.01 (0.81–11.25)	0.101
(Neo)adjuvant radiotherapy			
No	110	1	
Yes	24	5.05 (1.34–19.04)	0.017*
(Neo)adjuvant chemotherapy			
No	128	1	
Yes	6	3.71 (0.44–31.02)	0.226
Multivariate analysis			
Variable (n = 134)	No. of patients	Hazard ratio (95% CI)	p-value
Site			
Lower limb	79	1	
Upper limb	14	0	0.986
Trunk	41	6.28 (1.30–30.49)	0.023*
Tumor size (cm)	134	1.18 (1.06–1.33)	0.004*

*Statistically significant. CI, confidence interval

In this study, the occurrence of distant metastasis, but not local recurrence, was associated with an increased risk of tumor-related death in patients with a localized EMC. In a retrospective study of 41 patients with localized EMC, multivariate analysis revealed that local recurrence was the only factor associated with poor distant metastasis-free survival (HR 3.9 [95% CI: 1.1–14.7]; $p=0.04$) [14]. Patients with distant metastasis had poorer 10-year disease-specific survival than those without it (62% vs. 100%, $p=0.002$) [14].

Advanced stage extraskeletal myxoid chondrosarcoma

The distant metastasis rate after radical surgery for EMC is 25–50% [11, 13, 15, 19, 33–35]. However, survival is often prolonged even in the presence of metastatic

disease [11]. Systemic chemotherapy is necessary for patients with metastatic, unresectable, or progressive tumors [11]. Previously, EMCs were considered equivalent to conventional chondrosarcomas and were therefore considered refractory to the cytotoxic agents commonly used to treat sarcomas [18]. Few responses have been observed in patients with EMC treated with anthracycline, dacarbazine, or ifosfamide regimens [9, 13, 19–21]. However, in a series published by McGrory et al. [19], two of six patients with metastatic EMCs (33%) responded to multiagent chemotherapy. Additionally, a retrospective study of 10 patients with molecularly confirmed EMC collected by the Italian Rare Cancer Network showed greater sensitivity to anthracycline-based regimens than previously reported, with four (40%)

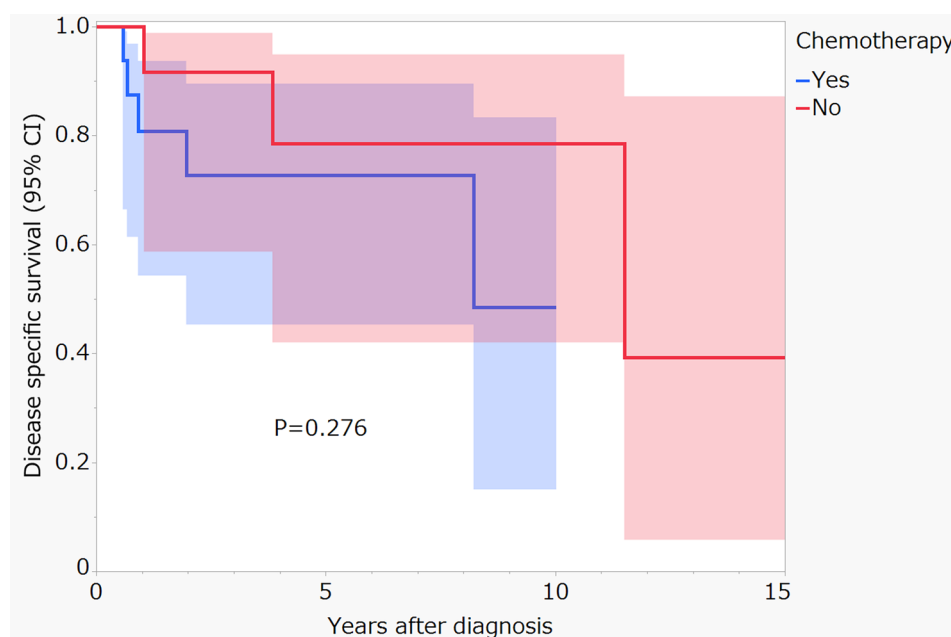


Fig. 3 Disease-specific survival rates of patients with and without advanced stage chemotherapy in patients with distant metastases at diagnosis. The shading around the curves represents the 95% confidence interval

partial responses and a median progression-free survival of 8 months [22].

Angiogenesis inhibitors are promising alternative systemic therapies with antitumor effects that are superior to those observed in other chondrosarcomas [23, 36, 37]. The first data came from a retrospective study of sunitinib, which reported six (60%) partial responses and a median progression-free survival of 8.5 months in 10 NR4A3-positive EMCs [36, 37]. This study was followed by a prospective single-arm Phase II trial of pazopanib in patients with NR4A3-positive advanced EMCs [23]. In total, 26 patients with EMCs were enrolled, and at a median follow-up of 27 months, four of the 22 evaluable patients had a partial response according to RECIST (all patients had EWSR1-NR4A3 fusion), with a median progression-free survival of 19 months [23]. A randomized phase III trial comparing trabectedin with the best supportive care in translocation-associated sarcoma showed stable disease for more than 1 year in patients with metastatic EMC [24]; therefore, trabectedin may also be considered. In this study, we found that the role of chemotherapy was limited in patients with localized and metastatic disease. However, because conventional chemotherapy was primarily administered to patients in this study, we cannot conclude whether new alternative systemic therapies are ineffective in patients with EMC.

This study had several limitations. First, as this was a retrospective study, there was an indication bias for (neo) adjuvant radiotherapy and chemotherapy use. Randomized controlled trials can avoid many of these biases by randomly allocating participants to groups. However,

because it is extremely difficult to conduct a randomized controlled trial for rare cancers such as EMC, the main objective of this study was to investigate prognostic factors. Second, although multivariate analysis was performed, the results must be interpreted with caution because of the small number of patients and events. Furthermore, the small number of patients may have led to type two errors. Significant factors may emerge as the number of patients increases. Third, there is no standard regimen for (neo)adjuvant, or advanced-stage chemotherapy. Future international collaborative studies should be conducted to increase the number of cases and to reevaluate the results of this study.

Conclusions

Patients with distant metastases at presentation had significantly shorter survival rates. In localized EMCs, the risk factors for local recurrence were inadequate surgical margins, and the local control effect of (neo)adjuvant radiotherapy was limited. The risk factors for distant metastasis were large tumor size, high histological grade, and inadequate surgical margin. The poor prognostic factors for survival were truncal origin and large tumor size. Chemotherapy had a limited effect on survival.

List of abbreviations

EMC	extraskeletal myxoid chondrosarcoma
HR	hazard ratio
IRB	institutional review board
IQR	interquartile range

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s13018-025-06245-6>.

Supplementary Material 1

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Author contributions

Conceptualization: TM, ST, and AN; methodology: TM, ST, and AN; formal analysis: TM, and ST; data curation: TM, ST, AN, and AK; writing – original draft preparation: TM and ST; writing – review and editing: CE; supervision: KH; and funding acquisition: ST.

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Data availability

The datasets generated and/or analyzed during the current study are not publicly available because of privacy considerations but are available from the corresponding author upon reasonable requests.

Declarations

Ethics approval and consent to participate

All procedures involving human participants were performed in accordance with the ethical standards of the institutional and/or national research committee and the 1964 Helsinki Declaration and its later amendments, or comparable ethical standards. The registry was approved by the IRB of the last author's institution (protocol code 2005-068).

Consent for publication

The IRB waived the requirement for informed consent because the data extracted from the database were anonymized.

Competing interests

The authors declare no competing interests.

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